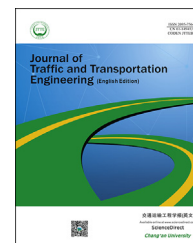


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Review Article

Review and evaluation of methods in transport mode detection based on GPS tracking data

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HIGHLIGHTS

- A systematic review of methods used to detect transport mode in GPS tracking data.
- Compare and evaluate the performance of the reviewed methods.
- Enhance knowledge for choosing algorithms to detect transport mode.
- Investigate the usefulness of additional GIS information to detect transport mode.
- Provide guidance for future research to develop transport mode detection.

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ABSTRACT

Mobility data, based on global positioning system (GPS) tracking, have been widely used in many areas, such as analyzing travel patterns, investigating transport safety and efficiency, and evaluating travel impacts. Transport modes are essential factors in understanding mobility within the transport system. Therefore, in this study, a significant number of algorithms were tested for transport mode detection. However, no conclusive recommendations can be drawn regarding which method should be used. The evaluation of the performance of the algorithms was not discussed systematically either in current literature. This paper aims to provide an in-depth review of the methods applied in transport mode detection based on GPS tracking data. The performances of the reviewed methods are then compared and evaluated to provide guidance in choosing algorithms for transport mode detection based on GPS tracking data. The results indicate that the majority of current studies are based on a supervised learning method for transport mode detection. Many of the reviewed methods first require manual dataset labeling, which can produce major drawbacks, such as inefficiency and human errors. It was also found that deep learning approaches have the potential to deal with large amounts of unlabeled raw GPS datasets and increase the accuracy and efficiency of transport mode detection.

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1. Introduction

The use of data, based on global positioning system (GPS), or some other equivalent position system, offers the possibility of following individual trips with regard to temporal and spatial positional characteristics in a much more detailed way, when compared to traditional travel surveys and travel diaries (Clifford et al., 2008; Nguyen et al., 2020; Rasmussen et al., 2015; Stopher and Shen, 2011). GPS tracking data are becoming ubiquitous in various transportation applications and have been widely used in collecting information on people and goods transport (Furletti et al., 2013; Gong et al., 2012; Sauerländer-Biebl et al., 2017). The advancements in GPS technology and devices used to collect data are improving rapidly, and data generated by GPS devices have become more accurate and reliable (Wu et al., 2016). In general, GPS tracking data provides information on longitude, latitude, date, time, speed, altitude, and direction of movement. In order to conduct further transport analysis, data processing is necessary to extract trip characteristics. The major processing steps are trip identification, map matching, and transport mode detection (Gong et al., 2014; Yang et al., 2015).

Many studies are available focusing on trip identification and map matching, while studies on transport mode detection (TMD) are still sparse (Huang et al., 2019). Only four review studies relating to transport mode detection based on GPS tracking data were found in the current literature, the latest being published in 2017. The limited number of reviews in the area could be due to the fact that transport mode detection based on GPS tracking data is challenging, especially when the data is unlabeled, i.e., there is no information regarding the transport mode used during a trip (Huang et al., 2019). Furthermore, while such reviews are rare and noncomprehensive, comparisons and

evaluations of the performance of the various methods are even more limited.

Therefore, this paper aims to address the outlined research gap by first conducting a systematic review of the methods applied for detecting transport mode based on GPS tracking data, and further by comparing and evaluating the performance of those methods. The following research questions will be the focus of the paper.

1. What kind of GPS data has been utilized for transport mode detection and specification of the datasets?
2. Which methods and algorithms have been used for transport mode detection, and their advantages and weaknesses?

The remainder of the article is structured as follows. In section 2, the method used for the literature review is presented. Section 3 reviews the methods applied in the literature to detect the transport mode. Section 4 compares and evaluates the performances of the reviewed methods. Section 5 concludes the study with highlights on the challenges in transport mode detection.

2. The method used for the literature review

2.1. Literature search strategy

In this study, we conduct a systematic review. The quality of a review depends on the scope and quality of included studies. In order to secure that the literature search ends up with publications of high quality in relation to the scope of the review, we have refined and reformulated the initial search strategy in an iterative process. PRISMA (Moher et al., 2009) is a protocol developed in order to secure the collection of high-

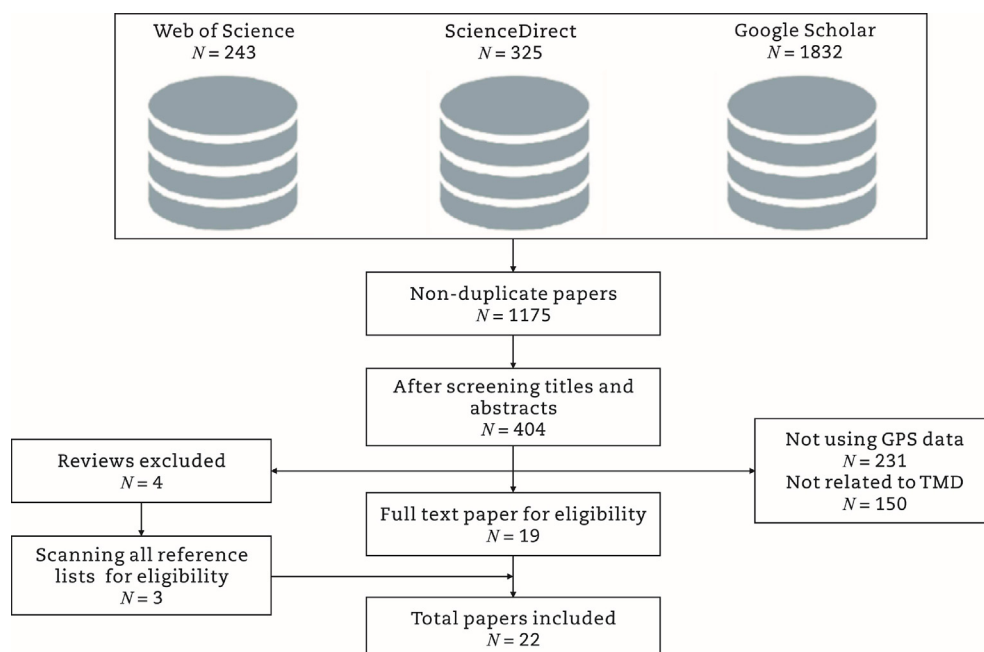


Fig. 1 – Flowchart of the systematic review process.

quality publications in relation to the scope of the review, and we follow that methodology in this study.

In the literature search, Google Scholar, ScienceDirect, and Web of Science are used to identify peer-review articles on transport mode detection based on GPS data. The literature search starts with three keywords, “transport mode detection”, “GPS data”, and “methods”. Additionally, the synonyms of these three keywords are applied in the search based on the three categories of keywords.

The first category of keywords includes the following synonyms: transport mode detection, travel mode detection, transport mode identification, and identify travel pattern. The second category includes machine learning, supervised learning, semi-supervised learning, unsupervised learning, statistical methods, and rule-based method. The third category of keywords comprises broad terms to ensure that the search does not miss relevant scientific work and includes the following terms: GPS data, GPS tracks, GPS processing methods, trip identification, trip recognition, GPS data cleaning, GPS trajectories, and map matching. The literature search was conducted in December 2019, and the metadata of each paper was collected. To ensure the eligibility of the scientific work for inclusion in the review, the articles were required to fit the following criteria: 1) the scientific work must be well written in English and published in a peer-reviewed scientific journal, conference proceeding, or book; 2) GPS tracking data must be used in the article; and 3) the article should focus on transport mode detection, not on other GPS data processing procedures. Fig. 1 shows the search and scanning process.

The number of articles collected from each respective database was: 243 from Web of Science, 325 from ScienceDirect, and 1832 from Google Scholar. After removing the duplicates, 1175 articles remained. The titles and abstracts of these articles were then screened; some articles were excluded in this process for the following reasons: 1) the article was not focused on transport mode detection; 2) it was a review article; 3) the full text was not available; and 4) GPS data was not used in the article. After this process, the number of articles to be further considered was reduced to 404. These articles were then read thoroughly. Nineteen of the articles focused on methodology for detecting transport mode using GPS data and were included in this review. In order not to miss related articles, the references list of previously excluded review articles in the screening check was rechecked, based on the PRISMA procedure. Finally, 22 articles in total matched all the criteria and were included in this review.

2.2. Findings from previous literature reviews on transport mode detection and their relation to this study

Before further discussing those 22 studies, we re-examined the four existing review studies mainly for two reasons. First, to compare the differences between the included studies and give an overall view. As shown in Table 1, Gong et al. (2014) had reviewed most studies up to the year 2013. Lin and Hsu (2014) only reviewed five studies. Wu et al. (2016) reviewed studies from 2008 to 2015, while Prelipcean et al. (2017) reviewed studies from 2005 to 2017. There was some overlap in these reviews, and the list helped us to check the validity of the paper to be included. Second, studying existing

studies shows the necessity for conducting a new review since more studies on the related topic were produced since 2017. It is necessary to update the review to provide more updated guidance and comprehensive knowledge in this area.

Gong et al. (2014) discussed alternative algorithms used to identify the transport mode. The reviewed analytical methods included rule-based methods, statistical methods (PM) (all the acronyms used in the paper are summarized and presented in Appendix 1), and machine learning algorithms (MLP, DT, BN, SVM, and CRF). As a complement to information available in GPS tracking data, Gong et al. (2014) stressed the importance of adding external data from GIS layers to improve transport mode detection accuracy. The rule-based method implemented in a GIS was found to be preferable. However, in this review, the analysis of how different analytical methods performed was limited.

Lin and Hsu (2014) focused on finding mobility patterns based on GPS tracking data. This study offered an overview of how the GPS dataset works and could be used in different research areas, such as location analysis, predicting movement, detecting transport mode, trajectory mining, and location-based activity recognition. The parts relating to transport mode detection were based on only four studies that covered statistical methods (probability matrix) and machine learning (supervised, unsupervised algorithms but not deep learning). However, the authors recommended unsupervised learning algorithms as preferable for transport mode detection. But, the comparison was based on only four studies, and the conclusion on the unsupervised algorithm was mainly obtained from just one study (Lin et al., 2013). The external validity of Lin and Hsu's study is, therefore, limited.

In the third study, Wu et al. (2016) reviewed 12 studies published during 2015 and focused on GPS data from cellphones. The studies reviewed mainly used a probability matrix, supervised, semi-supervised, and unsupervised machine learning, but no deep learning algorithm. The authors discussed how well the trip identification and segment identification made could significantly affect the performance of the transport mode detection. However, they did not conduct a thorough analysis of the performance of those reviewed algorithms. The study did not recommend which type of algorithm to use for detecting the transport mode.

Prelipcean et al. (2017) identified and provided a review of 22 papers in the area of transport mode detection. In 10 of these papers, accelerometers data, solely or in combination with GPS, were used. The main disadvantage of accelerometer data was they could not be used directly and were considered as an external data source complementing the information generated by the GPS device. The particular focus of this review was to analyze the difference between the three major research disciplines of transport science, location-based service, and human geography. The authors reviewed the statistical methodology and machine learning methodology (supervised algorithms) to detect the transport mode. They concluded that detecting the transport modes based on GPS tracking data had low accuracy since it was difficult to filter out the overlapping point. They also concluded that the use of external data, such as accelerometer data or GIS layer

Table 1 – An overview of the studies that have been reviewed in the four previous review papers and this review.

Reviewed study	This study	Gong et al. (2014)	Lin and Hsu (2014)	Wu et al. (2016)	Prelipcean et al. (2017)
Patterson et al. (2003)	✓	✓			
Stopher et al. (2005)		✓			
Chung and Shalaby (2005)					✓
Yuen et al. (2006)	✓	✓			✓
Byon et al. (2007)		✓			
Alvares et al. (2007)					✓
Reddy et al. (2008)		✓	✓		
Zheng et al. (2008b)	✓	✓	✓	✓	
Stopher et al. (2008)	✓	✓			✓
Palma et al. (2008)					✓
Bohte and Maat (2009)		✓			✓
Schuessler and Axhausen (2009)		✓			✓
Gonzalez et al. (2010)	✓	✓		✓	
Reddy et al. (2010)				✓	
Chen et al. (2010)	✓	✓			
Moiseeva and Timmermans (2010)		✓			
Zheng et al. (2010)			✓		✓
Stenneth et al. (2011)	✓			✓	✓
Zhang et al. (2011)				✓	
Gong et al. (2012)	✓	✓			
Lin et al. (2013)	✓		✓		
Biljecki et al. (2013)	✓				✓
Pereira et al. (2013)		✓			
Nitsche et al. (2014)				✓	
Byon and Liang (2014)				✓	
Yang et al. (2015)				✓	
Lari and Golroo (2015)	✓				
Rasmussen et al. (2015)	✓				✓
Xiao et al. (2015)	✓			✓	
Feng and Timmermans (2016)	✓				

Table 1 – (continued)

Reviewed study	This study	Gong et al. (2014)	Lin and Hsu (2014)	Wu et al. (2016)	Prelipcean et al. (2017)
Rezaie et al. (2017)	✓				
Sauerländer-Biebl et al. (2017)	✓				✓
Dabiri and Heaslip (2018)	✓				
Dabiri et al. (2019)	✓				
Yazdizadeh et al. (2019)	✓				
Li et al. (2020)	✓				
Markos and Yu (2020)	✓				
Yu (2020)	✓				

information, would improve accuracy in detecting the transport mode. This review discussed a rather limited number of analytical methods. It did not provide a comprehensive taxonomy and analysis of the performance of the analytical methodologies applied to detect transport mode.

In summary, one common finding from these four review studies is that pre-processing the data to extract essential information and using external information help increase the accuracy of transport mode detection. However, it is unclear from these review studies which algorithms were used to identify the transport mode. It is also unclear how well different algorithms perform in detecting the transport mode.

3. Methods for transport mode detection

Once trips are extracted from the positional recordings of the GPS, the data is ready to be analyzed in order to detect transport modes. In doing so, spatial and temporal information, such as trip features (e.g., speed, acceleration, trip route, trip distance) for each trip, is extracted. The features most used in detecting transport mode algorithms to assign a transport mode for each trip segment are then examined. In this section, we synthesize the results from the literature search. Tables 2–4 show the 22 selected articles and some critical features for detecting transport modes.

As can be seen in Table 2, among the three main methodological approaches (statistical methods, rule-based methods, and machine learning methods) for detecting the transport mode, machine learning is by far the most commonly used approach. Among the three main machine learning approaches in the more recent studies, it can be noted that semi-supervised and unsupervised machine learning were the most commonly used algorithms for identifying transport modes. This may be due to the fact that most GPS data are unlabeled; unsupervised learning can directly use the data, while supervised learning requires that the data is first fully labeled, which makes it time-consuming and inefficient. It can also be noted that deep learning emerges as the method used in recent studies. The use of statistical methodological approach is limited, and those studies are rather dated, while studies

using a rule-based methodology are more frequent over time; this is probably the case because rule-based methods are also developed for the use of unlabeled data. Studies are applying a rule-based methods used mainly two different algorithms, with no comparisons between them, while the statistical methods used only one algorithm.

Table 2 also shows that early studies mostly considered only one algorithm. However, a few articles tested up to ten algorithms. In total, 32 algorithms were tested (Table 3). Only DT, BN, RF, CNN, CA, and EL of these algorithms were evaluated and compared to other algorithms. SVM, MNL, C4.5, and MLP showed low accuracy, while the deep learning method performed with the least inaccuracy.

From Table 2, it can be seen that most studies detected between four and five transport modes. All studies detected the modes of walking, car, and bus, and many reported a detection accuracy of more than 90%. Notably, the structure of the datasets used in the studies varied, with some considering only one user and a few trips, while others considered 1000 users, and some recording movement for a few hours, while others spanned a period of years. In general, the resolution of the data was high, with positional recordings between 1 and 30 s. Also, the data source included collection from GPS devices, cellphone applications, and/or a combination of the two. The resolution of the dataset recorded by GPS devices was generally of higher quality than from the cell phone application. Studies using a combination of GPS devices dataset and cellphone application reported high accuracy in detecting the correct transport modes.

A description of input variables utilized in each study is listed in Table 4. All studies used speed as the primary variable in detecting the transport modes. The second most typical variable that applied in most of the studies (15 out of 22) was acceleration. All of the studies using the accelerometer found that acceleration was important in detecting the specific transport mode. With these two variables, it seemed possible to detect at least five transport modes with more than 80% accuracy. Besides these variables, other variables derived from the GPS devices or cellphones, such as trip distance and duration, heading change rate, and bearing rate, were also used. They were found especially helpful in detecting and distinguishing between transport modes, such as a car, public transportation, and bicycle. In many studies (nine), external data, such as GIS layers, road networks, and public transportation stations, have also been used to detect the correct transport mode.

4. Comparison and evaluation of the methods used in detecting transport mode

With the information acquired from the review, statistical methods, rule-based methods, and machine learning methods are found to be the three main methods used in detecting transport mode based on GPS tracking data. In this section, an in-depth analysis is conducted to evaluate which methods could be preferable by comparing the performances of the methods.

4.1. Statistical methods

A probability matrix (Stopher et al., 2008) is an algorithm that belongs to statistical methods for transport mode detection. Stopher et al. (2008) detected five transport modes (walk, bike, car, bus, train) using the probability matrix based on data from a GPS device. Several characteristics of the trip, such as average speed, maximum speed, most often recorded speed, and distance were used to define the probability of a specific transport mode. For instance, a walking trip with a high probability had an average speed of less than 6 km/h, while the maximum speed was not greater than 10 km/h. A final score was assigned to walking, bicycle, or motorized vehicles by summarizing the probabilities for a trip. After that, GIS layers information of the transport network was added to distinguish between a car and public transportation in a motorized category. Stopher et al. (2008) found that detecting bus trips separately from car trips sometimes was problematic, mostly when speed and GPS records were irregular. Compared to other studies, they used only speed as the primary variable in detecting the transport modes, and the remained transport modes were detected based on the external GIS information. In doing so, their probability model achieved a high accuracy (95%). Although Stopher et al. (2008) showed high accuracy, this study was problematic since the classification of transport modes was not reliable. This was mainly due to the GIS layer being added in the last step to distinguish between car, bus, and train, which could induce misclassification of the modes.

4.2. Rule-based methods

There are four fuzzy logic rules studies (Biljecki et al., 2013; Rasmussen et al., 2015; Sauerländer-Biebl et al., 2017; Yuen et al., 2006) and two studies of rule-based methods using GIS algorithms (Chen et al., 2010; Gong et al., 2012).

Yuen et al. (2006) also used data from a GPS device to detect four modes of transportation (walk, bike, car, bus), but using fuzzy logic rules instead. They proceeded in two steps: first, their algorithm only considered the GPS dataset; second, external data (transport network) were added and integrated into the analysis. By adding external data, the accuracy level increased above 90%. In this study, the fuzzy logic rules model is based on four variables (average speed, 95th percentile maximum speed, positive median acceleration, and data quality). The data quality variable was defined as the number of valid GPS records over the segment's total GPS records. In order to detect a transport mode, more than one rule from the rule base was used. A score was assigned to each rule by the fuzzy logic rules inference system. The set rules with the highest score were considered to best describe the transport mode on a segment. In their second step of the analysis, the integrated transit routes of the transport network improved the model's accuracy. However, compared to other studies, Yuen et al. (2006) used a small GPS dataset from only a few users with 109 trips and 237 trip segments.

Rasmussen et al. (2015) developed and combined fuzzy logic rules methods with a GIS-based algorithm to detect five transport modes (walk, bike, car, train and bus) by using GPS

Table 2 – Summary of methods in transport mode detection study.

Study	Main methodological approach				Accuracy (%)	No. of tested algorithms	Best algorithm	Detected transport mode	
	Statistical	Rule-based	Machine learning					Stationary	Walk
			Supervised	Semi-supervised					
Patterson et al. (2003)					✓	84.0	1	EM	✓
Yuen et al. (2006)		✓				91.0	1	FLR	✓
Zheng et al. (2008b)			✓			90.0	4	DT	✓
Stopher et al. (2008)	✓					95.0	1	PM	✓
Gonzalez et al. (2010)					✓	91.2	1	NN	✓
Chen et al. (2010)		✓				79.1	1	GA	✓
Stenneth et al. (2011)			✓			93.5	5	RF	✓
Gong et al. (2012)		✓				82.6	1	GA	✓
Lin et al. (2013)					✓	76.3	1	MD	✓
Biljecki et al. (2013)		✓				91.6	1	FLR	✓
Lari and Golroo (2015)			✓			96.0	1	RF	✓
Xiao et al. (2015)			✓			93.7	1	BN	✓
Rasmussen et al. (2015)		✓				90.6	1	FLR	✓
Feng and Timmermans (2016)			✓			99.6	7	BN	✓
Rezaie et al. (2017)				✓		>80.0	3	LP	✓
Sauerländer-Biebl et al. (2017)		✓				75.0	1	FLR	✓
Dabiri and Heaslip (2018)					✓	84.6	7	CNN	✓
Dabiri et al. (2019)				✓		79.8	9	CA	✓
Yazdizadeh et al. (2019)					✓	91.8	10	EL	✓
Yu (2020)				✓		91.5	6	EL	✓
Markos and Yu (2020)					✓	80.5	8	CAE	✓
Li et al. (2020)					✓	86.7	4	CNN	✓

Note: * means both transport modes are used,² means using the data from [Zheng et al. \(2008b\)](#).

data and an extensive set of GIS layers describing the multimodal transport network in detail. The GPS data consisted of 183 users of 752 identified trips over three to five days. Besides the sensitivity analysis, the algorithm was validated by comparing it with a control questionnaire collected among users. Their three-step model sought to detect the correct transport mode by applying the rail proximity layer in the first step and then identifying train trips, using speed, acceleration, and advanced analysis of GIS layer information. In the second step, fuzzy logic rules were applied to identify the modes of walk, bike, car, and either car or bus trips. In the last step, bus line alignment based on GIS layer information was used to separate bus and car trips. [Rasmussen et al. \(2015\)](#) were able to detect the five transport modes with an accuracy of 91%. However, the user data applied was still relatively limited, so that the results could be uncertain. Further, the methodology depended heavily on the use of GIS layers, which made the implementation and data processing time-consuming.

[Sauerländer-Biebl et al. \(2017\)](#) proposed a three-step algorithm to detect five transport modes (walk, bike, car,

bus, train) based on a small dataset collected from a mobile phone. In the first step, the trips were divided into segments by defining the end and beginning of the movement phase, based on the last trip information. In the second step, fuzzy logic rules were applied to differentiate between different modes. In the third step, the structure of subsequent stops in a trip was analyzed. The fuzzy logic rules were applied to detect the transport mode, including the input variables of turning angles, accelerations, duration of stops, and moving periods. The probability for a certain transport mode on a segment was set between 0 and 1. It was challenging to differentiate between car, bus, and train modes in a city, since the traffic rhythms (concerning speed, accelerations, and stops) for these modes were often rather similar. To solve this problem, the authors relied not only on accelerator data but also on additional indicators, such as distance, duration of the stops, and driver behavior, to identify recurring patterns. Compared to the previous studies, [Sauerländer-Biebl et al. \(2017\)](#) also used the change in turning angles after stops as a parameter. They found the use of turning angles after stops and acceleration to be

Detected transport mode									Input data	Resolution of the data (s)	User	Time period (d)	No. of trips
Running	Bike/e-bike	Motor-bike	Car	Bus	Train	Tram/metro	Aircraft	Boat/ferry					
			✓	✓					GPS device	2–10	1	0.5	29
	✓		✓	✓					GPS device	NA	9	58.0	109
	✓		✓	✓					GPS device	1	45	180.0	NA
	✓		✓	✓	✓				GPS device	1	NA	NA	NA
			✓	✓					Mobile phone APP	4	NA	NA	114
			✓	✓	✓	✓			GPS device	5	25–24	5.0–25.0	NA
	✓		✓	✓	✓				Mobile phone APP	15	6	21.0	NA
	✓		✓	✓	✓				GPS device	5	35	5.0	NA
	✓		✓	✓					GPS device	2–30	101	880.0	NA
	✓*		✓	✓	✓	✓	✓	✓*	Combining two datasets ²	5	1000	7.0	NA
			✓	✓					Mobile phone APP	NA	35	14.0	245,418
	✓*		✓	✓					Mobile phone APP	1	30	270.0	4685
	✓		✓	✓	✓				GPS device	1	183	3.0–5.0	752
✓	✓	✓	✓	✓	✓	✓*			GPS device	3	8	42.0–56.0	1554
	✓		✓	✓		✓			Mobile phone APP	NA	796	4154.0	NA
	✓		✓	✓	✓				Mobile phone APP	1	1	NA	NA
	✓		✓	✓	✓				Combining two datasets ²	1–5	69	1825.0	32,444
	✓		✓	✓	✓				Combining two datasets ²	1–5	182	1825.0	17,621
	✓		✓	✓					Mobile phone APP	NA	NA	NA	35,050
	✓		✓	✓	✓				Combining two datasets ²	1–5	182	1825.0	NA
	✓		✓	✓	✓	✓	✓		Combining two datasets ²	1–5	182	1825.0	17,621
	✓		✓	✓	✓				Combining two datasets ²	1–5	178	1825.0	25,995

beneficial in distinguishing between the modes of car, bus, and train. The result showed that the model could correctly detect 98% of 11 trips with a car in rural environments. However, the overall result showed that the model used in the study could only achieve 75% accuracy, which is considerably lower than other studies. The dataset was small and of poor quality. [Sauerländer-Biebl et al. \(2017\)](#) had one user conducting 11 trips, where each trip was more than 40 km for rural areas, which made it difficult to generalize the result.

Also implementing fuzzy logic rules methodology, [Biljecki et al. \(2013\)](#) identified ten different travel modes (walk, bike, car, bus, train, tram, metro, airplane, boat, ferry). They used both GPS device collected data and mobile phone collected data in combination with external GIS layer data. The data used were relatively large and comprised the movements of 1000 users over seven days with a positional recording for each second. The authors used time, longitude, latitude, and speed from the GPS devices and mobile phones. They also derived geographical information on railway, tramlines,

roads, bus lines, metro, and the location of the trajectory concerning water surfaces from GIS layers information. The core model was based on fuzzy logic rules using both GPS data and GIS information. In this study, the authors classified the movement trajectories based on a probability score, proposed by [Shortliffe and Buchanan \(1975\)](#) and re-named as certainty factor (CF) by [Biljecki et al. \(2013\)](#). This study developed the fuzzy logic rules based on the CF. For example, if the maximum speed on a trip segment was 118 km/h, they made a rule that assigned the transportation mode to be a car with a CF value equal to 1. If the maximum speed was higher on trip segments, the probability that it was a car was seen as lower, and consequentially the CF was set as lower than one for a car. In this way, the CFs for all transport modes in the fuzzy logic rules system were a function of available empirical evidence. In addition, [Biljecki et al. \(2013\)](#) also defined the speeds where the probability for a specific transport mode was equal to zero, and the CF was set to 0. In order to distinguish between bus, car, and tram, external data for bus and tram stops were used to set the CF.

Table 3 – Summary of used algorithm in all review studies with a focus on transport mode detection.

Algorithm	This study	Gong et al. (2014)	Lin and Hsu (2014)	Wu et al. (2016)	Prelipcean et al. (2017)
PM	✓		✓		
FR	✓	✓			✓
GA	✓	✓			✓
DT	✓	✓	✓	✓	✓
SVM	✓	✓	✓	✓	✓
BN	✓	✓	✓	✓	
CRF	✓	✓	✓	✓	
MNL	✓				
ANNs	✓				
RF	✓			✓	✓
MLP	✓	✓		✓	
NB	✓			✓	
LR	✓				
DTA	✓				
C4.5	✓				
KNN	✓				
RNN	✓				
CNN	✓				
GAN	✓				
DT-heuristic	✓				
CNN-A	✓				
Best CNN	✓				
Semi-two-steps	✓				
Semi-pseudo-label	✓				
SECA	✓				
CNN ensemble	✓				
LP	✓				
EM	✓	✓			
MD	✓		✓		
NN	✓	✓		✓	
Image-DNN	✓				
LSTM	✓				
DHMM				✓	✓
CNN-VC	✓				
DT-DHMM				✓	
CAE	✓				

Biljecki et al. (2013) showed that average and maximum speeds were essential variables in detecting the transport mode. They also found many overlapping characteristics between transport modes, and the accuracy was rather low when they did not use additional GIS layer data. The accuracy level of the model became 92% by adding geographical data as indicators. The authors reported that the accuracy level in detecting the correct transport mode of the model with a Gaussian combination was comparable with other machine learning methods. One limitation of this study is that there is no evaluation of the performance of the algorithm in relation to any other algorithm. Another limitation, compared with machine learning methods, is that fuzzy logic rules rely on many defined rules based on a specific scenario. The use of fuzzy logic rules requires much

human intervention compared to that required with machine learning methods.

The two studies (Chen et al., 2010; Gong et al., 2012) found in our literature search, using a rule-based methodology with GIS algorithms, were conducted by the same authors, using the same dataset. In the earlier study, Chen et al. (2010) proposed combining GPS data (such as longitude, latitude, date, time, number of satellites used, speed, and acceleration) with information from GIS layers (such as subway or commuter rail link, rail stations, bus line, and bus stops) to detect five transport modes (walk, car, bus, subway, and train). Chen et al. (2010) suggested a set of rules, such as maximum and average speed intervals, intervals of trip distance, and distances to public transport stops for detecting the transport mode. The authors used a multimodal transportation network to detect the transport mode in combination with the proposed set of rules. The reported overall accuracy level of this study was 79%. Among the five transport modes, the accuracy of the result reported ranges from 28% to 95%; accuracy in the detection of the travel mode of car was highest, while the accuracy level in detecting travel by train was lowest. In general, the accuracy levels in this study are the lowest compared to all the other studies included in this literature review. The main problematic part of the study by Chen et al. (2010) is the ability to correctly detect transit modes. However, the study area is very complex, comprising New York City, with several challenging environments for detecting the transport mode, including wide variability in density and diversity of land-use patterns, and a complex and vast transit network. The extremely low accuracy level in identifying the train as a transport mode is related to the low number of trips taken by train compared to other transport modes.

In the second study by Gong et al. (2012), the authors used the same datasets and an improved GIS algorithm to detect the same five transport modes as in the previous study by Chen et al. (2010). An area of improvement in this study was to expand the GIS algorithm to identify repetitive personal mobility patterns. This information was then matched with the characteristics of the transport network to detect the transport mode, which was also checked with a survey among the test persons. The overall accuracy rate increased to some extent compared to the previous study, ranging from 36% by train, to 92% for the walk mode. Compared to the first study, Gong et al. (2012) were able to increase the accuracy level by 10% for all public transport modes. However, in this study, the authors identified additional problems of relevance, such as traffic congestion and less accuracy in recording position during trips by subway or train. It was also proposed that the sophisticated transportation network of New York City with several infrastructure layers also added many difficulties to transport mode detection. The authors also found that the accuracy level in the mode detection was highly dependent on external data, such as GIS layers with information on the transport network.

Compared to statistical and machine learning methods, the criteria set by researchers are often based on ad hoc decisions, depending on the researchers' views and opinions, thus often resulting in much manual work.

Table 4 – Summary of input variable utilized in eligible study.

Study	GPS input variable	External input variable
Patterson et al. (2003) Yuen et al. (2006) Zheng et al. (2008b)	Velocity and a standard deviation of the velocity Average speed, 95th speed, positive median acceleration, the validity of GPS data Mean velocity, the covariance of velocity, top three velocities, and accelerations, length of the trips	Bus routes and stops Transport network
Stopher et al. (2008) Gonzalez et al. (2010)	Average, maximum, and most frequent speed, length of the trips Maximum and average of speed and acceleration, the ratio of the number of critical points over the total distance divided by time of the trip, total, and the average distance between critical points	Road and transportation network, ownership of a bicycle
Chen et al. (2010) Stenneth et al. (2011) Gong et al. (2012) Lin et al. (2013)	Travel time, speed, GPS signal quality Average of speed, acceleration, heading change rate and accuracy Average speed, 85th speed, duration, 85th speed, 95th acceleration Heading change rate, velocity change rate, stop rate, average velocity, an expectation of the velocity, variance of velocity, the first maximum velocity, and acceleration, the second maximum velocity and acceleration	Public transport network and bus stops and routes An average of a rail line, bus line, and closeness to bus stops Distance to the bus stops, train, and subway stations
Biljecki et al. (2013)	Speed, the 95th percentile of speed, mean speed, the mean moving speed	Railway, tramlines, roads, bus lines, metro, and the location of the trajectory concerning water surfaces
Lari and Golroo (2015) Xiao et al. (2015)	Delta bearing, accuracy, speed, delta speed, acceleration, delta acceleration Average speed, 95th of speed, average acceleration, low speed, travel distance, average heading rate	
Rasmussen et al. (2015) Feng and Timmermans (2016) Rezaie et al. (2017)	Speed, acceleration The standard deviation of speed, average speed, average acceleration, maximum speed, maximum acceleration, total distance Speed, trip duration, and trip length	Transport network Distance to the transportation networks, satellite information, ownership of a car, bike, and motorbike Distance between origin and destination and distance to the transit stops
Sauerländer-Biebl et al. (2017)	Mean and maximum speed, start acceleration change of heading after stop, active time since prior stop, length of deceleration and acceleration times, mean of the differences of the acceleration (horizontal and vertical movement)	
Dabiri and Heaslip (2018) Dabiri et al. (2019) Yazdizadeh et al. (2019) Markos and Yu (2020) Li et al. (2020) Yu (2020)	Speed, acceleration, jerk, and bearing rate channel Speed, acceleration, jerk, and bearing rate channel speed, acceleration, jerk, distance to the previous point, and bearing rate Speed, acceleration, jerk Speed, acceleration, jerk, and bearing rate channel Speed, acceleration, jerk, and turn rate time	

4.3. Machine learning methods

This section is divided into three subsections in which the performances and accuracy levels of the algorithms in the three main methodological approaches to machine learning are evaluated. Studies using deep learning are included in the part of the review where we discuss semi-supervised learning and unsupervised learning.

4.3.1. Supervised learning algorithms

In earlier studies, supervised learning methods were most commonly used for detecting the transport mode. These methods require all the data to be fully labeled, and for studies using this method the data processing steps are similar. In the beginning, the GPS data are divided into trip segments based on criteria for identifying a stop, usually when a positional recording has not changed for a certain amount of time. Several types of algorithms are used and tested in the reviewed studies, such as NN, BN, RF, MLP, SVM, DT, CNN, and KNN algorithms. These studies showed that the performance of the algorithms in themselves varies, depending on the number of input variables and transport modes, geographical context, as well as the resolution and size of the dataset.

Xiao et al. (2015) used four algorithms (SVM, MNL, ANNs, and BN) and showed that the BN algorithm performed more accurately than other algorithms (92%). Mobile phone applications collected the GPS dataset used in their study, and, in total, 1636 positioning data streams from 202 respondents on a daily basis were collected. After screening the data to ensure its entirety, 1248 data streams were stored for subsequent analysis. The authors used three rules for cleaning the data, the first being to remove unfinished positional recording that could show incorrect segments. The second rule was to remove positional recording with less than four satellites or with an HDOP of four or more, while the third rule was to remove positional recording with an altitude of higher than 200 m.

Feng and Timmermans (2016) applied and evaluated seven different types of supervised learning algorithms (BN, NB, LR, MP, SVM, DTA, and C4.5), and found that the BN algorithm showed the highest accuracy (99%). The GPS data used in this article were collected from a small group of people comprising eight individuals, carrying the GPS logger for six to eight weeks, with the device recording the position every 3 s. In total, 53,258 data points and 1554 trips were identified. Face-to-face interviews were conducted to record the individual's travel diaries, such as the timing and locations of activates of trips, transportation modes, and types of activities.

Both Xiao et al. (2015) and Feng and Timmermans (2016) applied attributes, such as average speed, maximum speed, average acceleration, and trip distance. Both studies indicated that the BN algorithm performed better than the other algorithms in detecting the transport mode. Feng and Timmermans (2016) also used external data and included attributes related to distance from transportation networks, satellite information, and ownership of the vehicle in order to detect the transport mode. They identified and labeled ten different transport modes (stationary, walking, running,

bike, motorbike, car, e-car, tram, bus, and train), and then applied seven supervised algorithms and evaluated their relative performance in detecting the transport mode. Results showed that the BN algorithm outperforms other algorithms in terms of accuracy. C4.5 and DT algorithms also had high levels of accuracy (98% and 95%, respectively); however, the dataset was rather small and consisted of only eight users conducting 1554 trips, which limited the external validity of the study. Unlike Feng and Timmermans (2016), Xiao et al. (2015) added the rate of movement and minimum speed as attributes, and identified five transport modes (walk, bike, e-bike, car, and bus).

Another algorithm used is RF, and a comprehensive evaluation of this algorithm can be found in Stenneth et al. (2011) and Lari and Golroo (2015). They evaluated and found that RF accuracy (91% and 93%) is higher than other supervised algorithms. Stenneth et al. (2011) detected six transport modes (stationary, walk, bike, car, bus, and train) from raw GPS data by applying the algorithms RF, NB, MP, DT, and BN. They combined GPS data collected by GPS devices for six individuals over three weeks with GIS information, such as transportation networks, bus stations, railway networks, and bus locations. The result showed the RF algorithm with a 93% accuracy in identifying different transportation modes. In addition to the GPS data, they also used GIS information, which improved accuracy by 17%. Besides the limited data size, there were two additional limitations related to the study by Stenneth et al. (2011). First, it did not consider trip identification for trip segments, and second, the number of training dataset was only six users. Lari and Golroo (2015) used GPS data collected from 35 users by mobile phone applications for over two weeks. More than 245,000 positional recordings were collected; variables of delta bearing, geographical position, speed, delta speed, acceleration, and delta acceleration were used to detect the transport modes of walking, car, and bus. The result showed that the RF algorithm achieved 96% accuracy. A limitation of the study by Lari and Golroo (2015) is that they used too few features to clean the data and remove the outliers (using only a maximum speed of 125 km/h for car and bus, and 20 km/h for walking) for the result to be reliable. Another limitation in their study is their choice of four different forests to check, where a scientific argument for this selection was not provided, which made the outcome of the evaluation challenging to interpret.

In another study, Zheng et al. (2008b) proposed an automated transport mode detection method, including three steps: an inference model, a change point-based segmentation method, and a post-processing algorithm based on conditional probability. In the inference step, the four algorithms applied were DT, BN, SVM, and CRF. To verify the effectiveness of their segmentation method, two methods are applied (uniform duration-based and uniform length-based segmentation). In other words, each segment had the same duration after being divided by the previous method, or the same length, after being divided by the next one. Their results also suggested that the DT algorithm achieves higher accuracy (90%) compared to the other three. It is somewhat difficult to evaluate this study's results since

it lacked a description of the raw data pre-processing steps, such as cleaning and recognizing the data, which may influence the accuracy of the segment and further decrease the accuracy of identification.

Of the five articles using supervised learning, two articles employed external GIS layer information of the transport network for detecting the transport mode (Feng and Timmermans, 2016; Stenneth et al., 2011). Most of the articles used average speed, variations of average speed, and acceleration as detection variables. Besides those GPS generated data, information on bearing rate and trip distance was also used. It can be concluded that average speed and acceleration are essential variables for detecting the transport mode. The use of external data, such as GIS-information, can help increase the accuracy level of the used algorithms.

4.3.2. Semi-supervised learning algorithms

Three relatively recent studies used semi-supervised learning algorithms. The results, though, are not congruent, and the studies concluded with three different best-performing algorithms. Rezaie et al. (2017) propose a semi-supervised algorithm based on the label propagation with KNN kernel to detect five transport modes (walk, bike, bus, car, subway) from GPS data collected by mobile phone application. They evaluated the accuracy of the KNN kernel compared to the two supervised methods of DT and RF. The results show that the accuracy of the proposed semi-supervised method outperformed the two supervised methods by slightly more than 80%. The running time of the semi-supervised was, on average, 16 times shorter than the supervised learning methods, and accuracy was more than 80%. The variables used were speed, duration, length of trip, origin, and destination. No external information was used.

Dabiri et al. (2019) tested three semi-supervised learning algorithms (two-steps, pseudo-label, and SECA), and six supervised learning algorithms (KNN, SVM, DT, MLP, CNN, and RNN). The dataset used in this study was part of the dataset collected by Zheng et al. (2008a). One of the semi-supervised algorithms (CAE) was developed by the authors and then tested against the other eight algorithms. Their aim was to detect the same modes as Zheng et al. (2008a), walk, bike, car, bus, with an additional mode of a train. To process the data, they first divided trips into single transport mode segments. The variables used in this study were the distance between two positional recordings, speed, acceleration, and jerk (the rate of change in the acceleration). The authors found that CAE outperformed the other algorithms and had an accuracy of 80%. The CAE algorithm combined two main components: a CNN classifier that worked with a labeled dataset, and a CAE that worked with both labeled and unlabeled datasets. The lowest accuracy level was for the walk mode, and the highest was for transport by train. One limitation of this study is the low number of users, with the labeled data only relating to two users, and the corresponding problem is that the algorithms did not perform well when the labeled data were scarce. Another problem is related to the need in CNN-based models to have a fixed-size input; the authors divided the

GPS trajectories of a trip into GPS segments with a fixed size. In the end, the fixed-length latent representation prohibited the model from analyzing trajectories with varying lengths, common in other available datasets.

More recently, Yu (2020) proposed DNN, a semi-supervised deep ensemble learning method, for transport mode detection when using a minimal number of labeled data for the task. He proposed a new deep learning DNN architecture, employing both the time and frequency domain of trips for transport mode detection. Based on the DNN architecture, a neural network consisting of four networks was constructed to generate proxy labels for unlabeled data, based on the knowledge of existing, but scarce, transport mode label information in the dataset. Yu (2020) used the same dataset as Dabiri et al. (2019) and Zheng et al. (2008a) to test the proposed algorithm, and also detected the same five transport modes as Dabiri et al. (2019). Yu (2020) further used the same variables as Dabiri et al. (2019), with one exception, the use of turn rate time instead of bearing rate. To evaluate the performance of the proposed algorithm, he compared it with a selection of algorithms used by Dabiri et al. (2019) and Zheng et al. (2008a). The result showed that the accuracy of the algorithm developed by Yu (2020) outperformed the algorithms tested by Dabiri et al. (2019). Yu (2020) argued that the accuracy of the GPS position did not have a notable influence on DNN algorithm performance. Further, sensitivity tests of features showed that all data features employed by the algorithm contributed to the identification task with different importance.

When all segments are not labeled, the semi-supervised algorithms perform better than supervised algorithms. Further, if only a sample of the data is labeled using a semi-supervised algorithm, the accuracy levels are still at the same level as the supervised algorithm that used a fully labeled dataset. This clearly shows that there is a limited need to label the data before identifying the transport mode. The advantage of semi-supervised algorithms compared with supervised learning methods is that a large volume of data can be analyzed with fewer efforts since the need to label the data is much less. We did not find that the GIS layer was used in the studies evaluating semi-supervised methods. However, applying such external data can probably improve the accuracy level in detecting the transport mode.

4.3.3. Unsupervised learning algorithms

Unsupervised algorithms used to detect transport mode were applied in five of the reviewed articles (Dabiri and Heaslip, 2018; Gonzalez et al., 2010; Li et al., 2020; Markos and Yu, 2020; Yazdizadeh et al., 2019). In three of these, a deep learning algorithm was applied. In the other two articles (Lin et al., 2013; Patterson et al., 2003), a simpler algorithm was applied, where the transport mode of the unlabeled data was detected through a clustering method based on key attributes.

In an early study by Patterson et al. (2003), expectation-maximization was used, and speed was considered as a critical feature for clustering the data. Three transport modes (walk, bus, and car) were identified in this study, and the accuracy was 84%. The dataset was small, with only one

user wearing the GPS device for 24 h. The authors also used external GIS data to detect the transport mode. They argued that the low accuracy level (84%) depended on the algorithm's inability to differentiate between transport modes with movement at low speed.

Lin et al. (2013) proposed the MD algorithm that relied on Kolmogorov-Smirnov. Their implementation of the MD algorithm relied only on instant speed, average speed, and maximum speed. The dataset was rather large and consisted of 101 users wearing a GPS device for 29 months. Four transport modes were detected (walk, bike, car, and bus). Using the MD algorithm resulted in an accuracy of 76%, and the performance was not tested against other algorithms. In their application, the authors first assumed that the speed distributions on the street segments are similar for different vehicles of the same transport mode, and second, the speed distributions between modes varied, especially in the high-speed part of the speed distributions. They found that the challenging part in detecting the correct mode was at the higher end of the speed distributions for each transport mode.

Gonzalez et al. (2010) identified three types of transport modes (car, walk, and bike) by considering the critical points of the trips. In this study, the maximum and average rate of acceleration and velocity, the total and average distance between critical points, and the ratio of the number of critical points over time and distance were used in the detection process. The authors used NN to detect travel modes, and mobile phones to collect data for an unclear number of users and time. In this research, two types of datasets, all GPS points and a selected sample of critical GPS points were used. The data consisted of 114 trips, indicating that it was rather small. In order to train and test the NN algorithm, the authors applied 10-fold cross-validation, and the highest accuracy was 91%. Their study has two potential problems. The first limitation is that only 79 trips are used for training, which is rather small to be reliable, and secondly, the users manually labeled the dataset. Both of these potential problems might affect the accuracy of mode detection. Another limitation is that it only considered three types of transport mode.

In a recent study by Yazdizadeh et al. (2019), a CNN algorithm was developed to detect four modes of transport (walk, bike, car, and public transport) based on a GPS dataset collected by mobile phone application. The data consisted of 35,000 trips. However, the number of users' resolution of the data and time period for the data collection were unclear. The authors identified the segments based on the trip-breaking algorithm, considering dwell time between GPS positional recording. The trip-breaking algorithm detected trip segments based on 3-min gaps in the data. The applied feature in this study was the same as Dabiri and Heaslip (2018), and it also used turn rate time, as used by Yu (2020). They considered the distance to previous positional points. In their final model, Yazdizadeh et al. (2019) combined the result of the CNN algorithm with average and majority voting and optimal weight methods. The obtained results exploited the ensemble library by applying an RF model as a meta-learner. The ensemble method, with the RF as a meta learner, resulted in

an accuracy of 92%, which, compared to the performance of algorithms considered in this review, is at the top level.

Manually labeling data takes time and potentially creates many problems in achieving correct labeling. Dabiri and Heaslip (2018) used CNN schemes to automatically derive high-level features from the raw data input. They identified the transport modes (walk, bike, car, bus, train) using the same dataset as Dabiri et al. (2019) and Zheng et al. (2008a). In total, they tested seven algorithms, of which two were unsupervised deep learning algorithms (CNN-A and Best CNN), and five classical supervised learning algorithms (KNN, SVM, DT, RT, and MLP). Without using labeled data or external data, they found that CNN performed best with an accuracy of 85%. Using the same dataset and features as Dabiri and Heaslip (2018), but developing another deep learning algorithm, Yu (2020) outperformed their accuracy level in detecting the transport mode. In addition, Yazdizadeh et al. (2019) used the same algorithm as Yu (2020) and reached the same accuracy, but the dataset used in this study was different.

Unsupervised deep learning algorithms have been used in recent studies by Li et al. (2020) and Markos and Yu (2020). The applied dataset in both studies is the same as Dabiri and Heaslip (2018), Yu (2020), and Zheng et al. (2008a). The main contribution of the study proposed by Li et al. (2020) is the use of a GAN algorithm to artificially create training samples data to increase not only the required sample size but also to improve the accuracy of the detection model by balancing the dataset. When the CNN algorithm is applied to the balanced dataset in order to detect the transport mode, the accuracy level of the model increases to 86%. However, by using GAN to create a more trained and balanced dataset, the accuracy of the model is improved for identifying the bus and car modes. In another study, the CAE algorithm is considered for detecting the transport mode (Markos and Yu, 2020). However, the accuracy level of the model in this study compared to the other studies is low (80.5%), but the applied dataset is fully unlabeled in this work.

The deep learning method uses raw data rather than a set of hand-crafted features, which is a significant advantage compared to classical machine learning algorithms. Further, from the studies by Dabiri and Heaslip (2018), Dabiri et al. (2019), Yazdizadeh et al. (2019), and Yu (2020), it can be concluded that deep learning algorithms perform better with higher accuracy in detecting transport mode compared to the more simple, unsupervised machine learning algorithms discussed above. However, Li et al. (2020) indicated that the accuracy can be improved by using the GAN algorithm to have a balanced dataset and increase the accuracy level of the model to detect the correct transport modes. Another deep learning algorithm that can be used in transport mode detection studies is the GCN algorithm that has been widely used in traffic and transportation prediction. The GCN algorithm can be a good candidate to be used in transport mode detection using GPS data. Thus, deep learning has the potential to introduce new and more efficient ways of processing raw data in transport mode detection.

5. Discussions and conclusions

This paper presents a review of existing research using different methods in detecting transportation modes based on GPS tracking data. We have found that mainly three categories of methods have been used: statistical methods, rule-based methods, and machine learning methods. Among those methods, various algorithms have been tested in different contexts. It is found that machine learning methods, including supervised, semi-supervised, and unsupervised learning algorithms, are more often used than the other two methods. The review also focuses on comparing and evaluating the applied algorithms to provide further guidance for transport mode detection. In the evaluation, we highlight the structure of the used datasets, implementation of the algorithms, use of external data, and accuracy of the detection results.

The results show that three to five modes are the most common number of modes to be detected, while some studies have attempted to detect as many as 12 transport modes. The most common and powerful variables used to detect transport mode are average speed, maximum speed, and acceleration. It is also evident that using external information can increase the performance of the used algorithms. The drawback of using more GPS generated variables, external data, and labeled data is time-consuming in organizing the information and conducting the computation, which can result in inefficiency. However, when it is possible to label the data, it improves the accuracy level of the algorithm.

The findings in this study show that there is no standard algorithm used for transport mode detection. It is currently difficult to suggest any best method since the dataset, scenarios, and purpose of applications vary. Further, the evaluation of the performance of algorithms suffers from a lack of standard procedures. However, using some datasets as benchmarks could be helpful in finding the best algorithm in a specific transport mode detection. Although one dataset exists (GeoLife) that has been used on some of the reviewed studies (Asci and Guvensan, 2019; Dabiri and Heaslip, 2018; Markos and Yu, 2020; Zheng et al., 2008a), it would be good for the future research to develop more dataset that could be used as benchmark dataset.

In this study, the advantages and disadvantages of statistical, rule-based, and machine learning methods have been analyzed. With regard to the statistical methods, we found that a disadvantage is that it has not shown to be able to distinguish between modes like a car, bus, and train with high accuracy. An advantage with the rule-based methods when using GIS layers information of the transport network is that it works well in identifying as many as 12 transport modes. The drawback of this model is that it relies on the existence of some prior understanding of the detected transportation modes. This is problematic since it requires many manual tasks for defining rules that rely on researchers' opinions, besides being time-consuming, and it is also difficult to know how well the data is labeled. The machine learning methods

have the major advantage of not needing human intervention since they rely on established statistical clustering techniques. However, a drawback to a varying degree with these methods is that they also need labeled data. Especially for supervised learning algorithms, that is problematic since all data fed into the algorithm needs to be labeled. This limits the use of supervised learning algorithms to be applied on a rather small dataset. The problem of labeling for the data still remains for semi-supervised learning algorithms, although it is not that massive since only a proportion of the data needs to be labeled. For the unsupervised learning algorithms, this is not a problem since they work with unlabeled data. The advantage of a supervised algorithm is that the accuracy is higher compared to all other machine learning algorithms. A disadvantage for the unsupervised learning algorithm is that the clusters that the analysis ends up in can be difficult to trace to specific transport modes, especially when the differences between clusters are very small. In this way, the unsupervised learning algorithms share the problems that exist for the statistical methods.

Some recently made studies have used deep learning algorithms applied to the transport mode detection problem. Deep learning has been found promising in achieving a high accuracy level (above 90%). However, current studies are limited, and more studies with larger datasets are needed for evaluating deep learning algorithms. Compared with other methods, semi-supervised learning with small proportions of labeled data and unsupervised learning, in combination with deep learning algorithms, could be good methods for providing more efficient and accurate results.

In addition, common limitations in many of the reviewed studies are the small datasets and lack of systematic standardization in collecting data, and features used to clean the data and detect transport modes. Improving data collection methods and validation techniques for large-scale mobility data ought to be a focus of future studies. Last, to enhance state of art in transport mode detection, research works are still needed, especially in: 1) improving algorithms that are able to distinguish between transport modes, particularly those often driven with similarity in speed, and 2) providing benchmark datasets to make it easier to compare usability and the performance of different transport mode detection methods.

Conflict of interest

The authors do not have any conflict of interest with other entities or researchers.

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Appendix A

Table A1 – Nomenclature for common transport mode detection algorithms.

Abbreviation	Description
ANNs	Artificial neural networks
BN	Bayesian network
CA	Convolutional autoencoder
CAE	Convolutional auto-encoder
CNN	Convolutional neural network
CNN-A	Convolutional neural network architecture
CNNE	Convolutional neural network ensemble
CRF	Conditional random field
DHMM	Discrete hidden Markov model
DT	Decision tree
DTA	Decision table
DT-DHMM	Decision tree discrete hidden Markov model
EL	Ensemble learning
EM	Expectation maximization
FLR	Fuzzy logic rules
GA	GIS algorithm
GAN	Generative adversarial network
GCN	Graph convolutional network
GIS	Geographic information system
HDOP	Horizontal dilution of precision
Image-DNN	Image deep neural network
KNN	K-nearest neighbors
LP	Label propagation
LR	Logistic regression
LSTM	Long-short term memory
MD	Modetect
MLP	Multilayer perception
MNL	Multinomial logistic regression
NB	Naïve Bayesian
NN	Neural network
PM	Probability matrix
RF	Random forest
RNN	Recurrent neural networks
SECA	Semi-supervised convolutional autoencoder
SVM	Support vector machine
TMD	Transport mode detection

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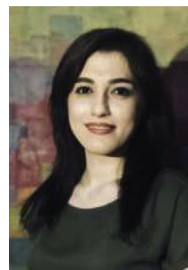
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